

Skills

Languages: C++, C, MATLAB, Lua, $\LaTeX$

Software: Git, ROS2, OMPL, DDS, Qt, Simulink, FreeRTOS, Autodesk Inventor (Certified), SolidWorks, Rhino

Hardware: PCB Design, Breadboarding, Computer Architecture, Oscilloscope, Multimeter, 3D-Printing

Technical Experience

Autonomous Systems Engineer

Aug 2025 – Present

Georgia Tech Research Institute • Applied Systems Lab

Atlanta, GA

- Developed C++ software for collaborative autonomous UAV swarms.
- Git GNU/Linux and CMake C++14
- Software systems integration
- Embedded software development
- Refactored legacy code to use DDS middleware.
- Autopilot
- Integrated and benchmarked custom path planning algorithms using The Open Motion Planning Library (OMPL) and GeographicLib into collaborative autonomy software.
- Implemented track association algorithms for multi-object tracking.

Robotics Research / M.S. Computer Engineering

Aug 2023 – May 2025

Virginia Tech • Control, Optimization, and Online Learning for Autonomy Lab

Blacksburg, VA

- Built a 6-DOF robotic arm and an accompanying ROS2-Gazebo simulation environment for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.

Thrust Vector Control Intern

May 2024 – Aug 2024

Jacobs Space Exploration Group • Thrust Vector Control Test Lab

Huntsville, AL

- Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center.
- Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.

Control Research Intern

Jun 2023 – Aug 2023

G2ELab • Power Electronics Lab

Grenoble, France

- Researched different inverter control methods for 4-leg (microgrid) topologies in Simulink and HIL simulation.

Naval Research Enterprise Internship Program

Jun 2022 – Aug 2022

Naval Surface Warfare Center • Carderock Division

Bethesda, MD

- Developed concept hospital sea-train designs, estimated fuel consumption, and electrical power loads.

Projects

6-DOF Robot Arm

- 3D printed robot arm, built using stepper motors and pulleys.
- Implemented ROS2 Jazzy control and Gazebo Harmonic simulation.

Education

M.S. in Computer Engineering

May 2025

Virginia Tech • Bradley Department of Electrical & Computer Engineering

Blacksburg, VA

- Focus: Software and Machine Intelligence
- Research Group: Control, Optimization, and Online Learning for Autonomy Lab at UT Austin

B.S. in Electrical & Computer Engineering (Double Major)

May 2024

Virginia Tech • Bradley Department of Electrical & Computer Engineering

Blacksburg, VA

- Focus: Controls, Robotics, and Autonomy
- Design Teams: SolarCar at VT and Human Powered Submarine